

REVIEW

Demystifying the development of dendritic cell subtypes, a little

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The broadest definition of dendritic cells (DCs) is white blood cells that can take up antigen, process it and then present antigen-derived peptides to activate cognate naive T cells. Although this definition is by no means perfect, it is nevertheless now textbook. The source of frustration more recently has focused on other issues, including the distinction of the DC subtypes, their differential roles in the immune system, their lineage relationship to each other (and other leukocytes) and whether the mouse and human DC findings overlap. Here, I condense the classification of DCs in both the steady state versus infection, with primary focus in the mouse. Emphasis is then given to debates surrounding the *in vivo* pathways of DC differentiation in different conditions, which culture models best represent these processes (*fms*-like tyrosine kinase 3 ligand versus granulocyte-macrophage colony-stimulating factor), and what the human and mouse DC subtype equivalents might be. In addition, a model termed 'graded' commitment is proposed that, as a departure from the classic binary models of hematopoiesis, attempts to explain the recent clonal data where subtype-specific DC precursors branch from this pathway.

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Prior to the 1970s, lymphocytes were regarded as an indistinguishable group of cells, primarily because they appeared identical under the light microscope. In fact, when Mitchell and Miller first wielded the axe separating them into T versus B lymphocytes, they were ridiculed as 'complicating the field'.^{1–3} In a case of history repeating, a similar sentiment was given to those whom first identified the existence and properties of the dendritic cell (DC) subtypes^{4–6}—an exercise that was often referred to ignominiously as 'stamp collecting'. However, arising from the early work of Shortman, Steinman, Moser, Hart, MacPherson and their colleagues, it is clear that DCs are not just the one cell type simultaneously performing the entire repertoire of professional antigen-presenting cell (APC) functions.

On the contrary, the DC subtypes appear to be distinct entities with different life histories. They differ not only in phenotype, but also in their genetic program, anatomic location, expression of receptors for pathogen molecules and abilities to influence T-cell fate (reviewed in Shortman and Naik⁷ and Villadangos and Schnorrer⁸). This specialization and division of labor presumably allows the immune system to intricately 'tailor' an adaptive immune response appropriate for the breaching pathogen—like having different generals for the army, navy or air force. Yet despite clear differences in the DC subtypes and their functions, evidence of their unique origins has only recently been shown.

This review attempts to introduce the immune system's putative need for DC subtypes, briefly introduce the players, summarize some aspects and controversies surrounding their development, investigate

the different culture models of DCs, introduce a new model to explain branching of DCs at various stages of hematopoiesis and address the potential human counterparts of the mouse DC subtypes. As this review cannot be exhaustive, the reader is directed to a past thesis available online for some further information.⁹

WHY SO MANY SUBSETS?

It may seem to those purveying the field that a new DC subtype is discovered almost seasonally. The question is often asked, 'Why do models of the immune system require yet more DC subtypes?' The question should instead be, 'The DC subtypes do exist, and appear to have different functions—why might this be important?' Otherwise, one potentially follows the same cynicism of those whom discredited the T- versus B-lymphocyte distinction. Granted, some phenotypic differences mean little in the context of differentiation or function. However, in other cases, such differences correlate to fundamentally distinct cell types whose functions are worthy of unraveling—for both understanding of the immune system, and for potential clinical benefits.

As organisms became more complex throughout evolution, so too did their immune system in order to better defend themselves, and their ability to 'play host' to a greater range of microbes. Accordingly, only since the emergence of the jawed vertebrates did the adaptive immune system develop the important cytotoxic T and antibody-forming B lymphocytes.^{10,11} The processes that link the innate immune detection of pathogens to the instruction of the adaptive

immune response appear to have developed over evolutionary time.¹² DCs appear to be one of the most important of these processes and they must contend with several variables when providing such a link. These include (1) the distinct kinds of pathogens, (2) distinct routes of infection and (3) the inhibitory mechanisms employed by pathogens to limit their detection.

How do DCs sense infections? Pathogens unwittingly, or at least unwantingly, signal their presence by their associated molecular patterns or resulting pathology,¹³ for which DCs, in turn, have pattern recognition receptors.¹⁴ If this is the case, could not the one kind of DC conceivably arm itself with all receptors and respond only in accordance to each infection? The answer could be 'yes', and one might find an example if lower-order animals were probed for such a putative ancestral DC subtype. But at least in many mammalian species, the answer appears to be 'no', with the presence of many distinct and overlapping DC subtypes.^{7,15–18} Perhaps ancestral DCs split at some evolutionary stage, and the advantages of a division of labor of multiple DC subtypes outweighed the simplicity of having just the one—akin to the split in T versus B lymphocytes that occurred in the transition from jawless to jawed vertebrates.^{12,19} This is speculation and the issue is better suited to the dedicated immune evolutionist.

DC CLASSIFICATION

As previously described, DCs can be categorized by some general properties of their life history.⁷ Our group has preferred to classify the DC family by the following criteria (upon which I have elaborated here somewhat); first, either as precursors of DCs (pre-DC) or the DC products themselves; second, according to whether the DCs migrate from the peripheral tissues via the lymphatics to the draining lymphoid organs ('migratory'), develop and reside within the lymphoid organs ('resident') or develop in the bone marrow (BM) and are circulating in the bloodstream ('circulating'); third, whether they are present in the uninfected steady state or whether they arise with infection or inflammation ('inflammatory'); fourth, into the different subtypes and fifth, whether the DC itself is 'unactivated' or 'activated' (the latter could result in T-cell activation or tolerization) (Table 1). Each of these classifiers could, in turn, be given more levels of complexity (for example, which organ they are resident in, different types of activation and so on), but for the purpose of this review it will be left there. Any cell of the DC family, at any one time, should be amenable to a value for each of the five classifiers described above.

Pre-DCs versus DCs

Cells in the immediate previous step of DC development are classified as pre-DCs. They can differentiate into DCs by default in the steady state, or may require additional signals only provided by infection or inflammation.⁷ This is an important distinction to make as some pre-DCs, such as monocytes and plasmacytoid cells (pDCs), also have unique functions in the steady state, independent of their capacity for DC differentiation (discussed later), and are circulating. Nevertheless,

their potential as pre-DCs ascribes them a property to distinguish them from non-DC APCs such as B cells and macrophages. Other pre-DCs, such as pre-conventional DCs (pre-cDCs) develop into DCs by default in the steady state (also discussed later). 'Developed' DCs, by contrast, are 'dendritic' when freshly isolated, even though there may still be differences in levels of 'maturation' or immunogenicity (see below).

Migratory

Migratory DCs are 'born' in the periphery, act there as sentinels for pathogens or peripheral self-antigen, then migrate via the lymphatics toward draining lymphoid organs. This process is usually accompanied by a kind of maturation and occurs at a basal rate in the steady state^{20–22} so that they are found in the draining lymphoid organs as 'mature' DCs.²³ Surprisingly, this process can occur even in germ-free or Toll-like receptor (TLR) signaling-independent scenarios,^{24,25} but can also be enhanced upon infection or inflammation.^{26–30} Thus, crucial differences other than major histocompatibility (MHC) class II levels, a classic activation marker, differentiate mature steady-state DCs versus mature pathogen-activated DCs. Factors can include co-stimulatory molecules and cytokine production that must always be taken into account when determining any influence on T-cell fate.³¹ Langerhans cells (LCs) and dermal DCs fit into the migratory DC category.

Resident

Lymphoid-tissue resident, or 'simply resident DCs', are located in all lymphoid organs of the mouse, including the spleen, and are found in an MHC II^{int} 'immature' state,²³ which allows their clear distinction from the mature MHC II^{hi} migratory DC.²¹ Resident DCs include the CD11c^{hi} CD45RA^{lo} MHC II^{int} cDC, which can be further broken into two broad subsets; the CD8⁺ cDC and the CD8[−] cDC.^{5,6} Most CD8[−] cDCs also express CD4 but are largely overlapping in function with those that do not.^{32,33}

Circulating

Although it is clear that some pre-DCs, including pDCs and Ly6C^{hi} monocytes, can circulate in blood and extravasate tissues depending on the circumstances, one should take care not to refer to these as circulating DCs. Some arguments favor that cDC migrate via the blood and can seed the spleen and lymphoid organs.^{34,35} However, CD11c^{hi} cDC are actually not found in significant numbers in the blood,^{36,37} and only some in BM.³⁸ The majority of CD11c⁺ cells in the blood are B220⁺ pDCs, DX5⁺ NK cells, MHCII[−] pre-DCs³⁹ or Ly6C^{lo} monocytes (which might also overlap with blood pre-DC1).^{37,40,41} No cell freshly isolated from mouse blood has, to my knowledge, been demonstrated to be a good T-cell activator without prior hematopoietic development.³⁷ Why this conflicts with the observation of DCs in human blood¹⁸ is unknown, but might be explained by 'dirty' humans versus 'clean' mice, or indicate a species difference.

THE PLAYERS

Monocytes

Monocytes can be pre-DCs, and are phagocytic cells of the myeloid lineage that are found in several organs, including blood, BM and spleen at low levels (Figure 1). Upon infection or inflammation, they are recruited into the bloodstream and organs for innate immune functions^{42,43} (Figure 2). They represent a relatively terminal stage in what is often referred to as the mononuclear phagocyte system, or MPS⁴⁴ and probably derive from macrophage-DC precursors

Table 1 DC classification

1	DC stage	Pre-DC	Developed DC	
2	DC life history	Migratory	Resident	Circulating
3	Host state	Steady state	Inflammatory	
4	DC subtype	The different DC subtypes		
5	DC activation state	Unactivated	Activated	

Abbreviations: DC, dendritic cell; pre-DC, precursors of DC.

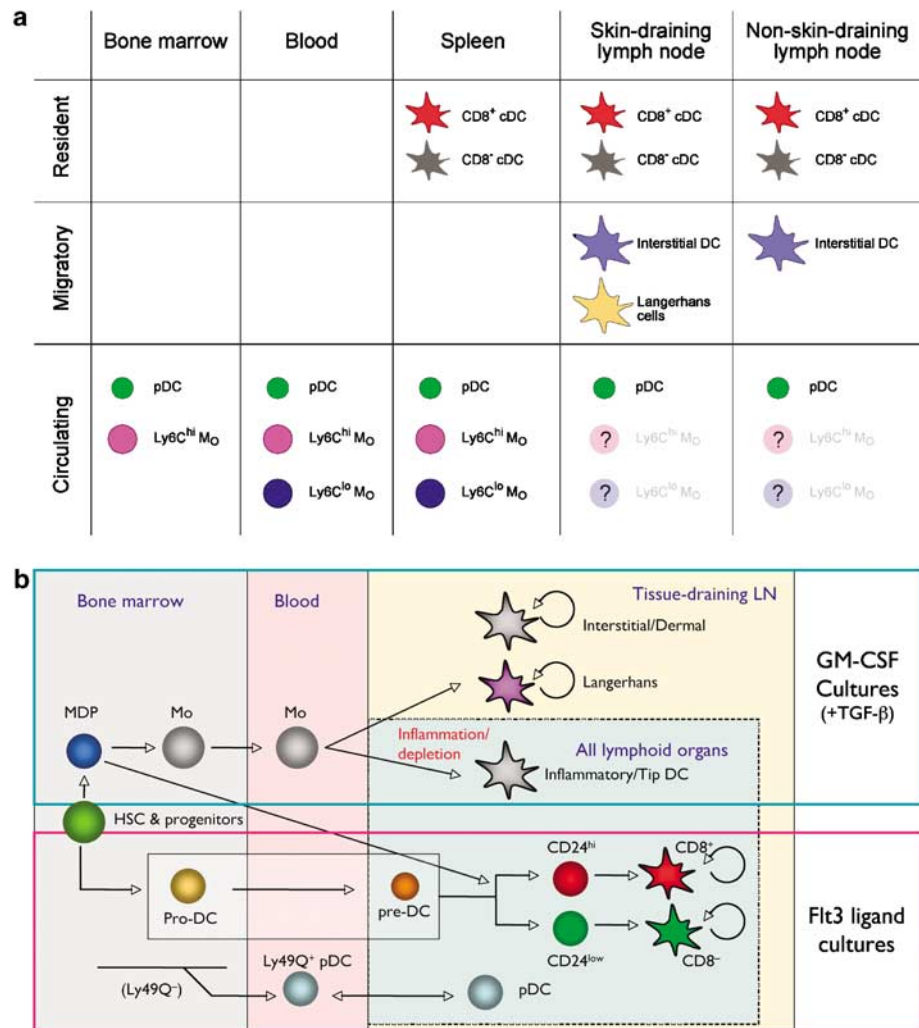


Figure 1 Location and developmental pathways of the dendritic cell (DC) subtypes. **(a)** The location of precursors of DC (pre-DC) and DC subtypes in a steady-state mouse. **(b)** Hematopoietic stem cells (HSCs) and their progenitors—including common myeloid progenitor (CMP) and common lymphoid progenitor (CLP)—give rise to more committed progenitors, including the macrophage-DC precursor (MDP) and pro-DC. MDPs give rise to both monocyte subtypes, which can differentiate into DCs. This latter process occurs at a low to absent level in the steady state, and is enhanced with inflammation. Development of these DCs can be recapitulated in bone marrow (BM) cultures with granulocyte-macrophage colony-stimulating factor (GM-CSF) ($\pm$ IL-4, $\pm$ TGF- β). Pro-DCs (CD11c⁺MHCII⁺) divide and differentiate into pre-DCs (CD11c⁺MHCII⁺) *en route* to generation of the DC subtypes, and this can be mimicked in *fms*-like tyrosine kinase 3 (flt3) ligand BM cultures. As both of these cell types are found in BM and lymphoid organs, it is not clear where this differentiation step takes place. In any case, pre-conventional DCs (pre-cDCs) in spleen have committed precursors to either CD8⁺ cDC only (CD24^{hi} pre-cDC) or CD8⁺ cDC only (CD24^{lo} pre-cDC). Some, but not all, DCs are capable of low-level homeostatic self-renewal in the steady state. Most splenic DCs are replaced by blood precursors within 10–14 days, whereas dermal and Langerhans cells (LCs) are stable and not replaced. The steady state changes in cases of inflammation, infection or DC loss. In these circumstances, monocytes can be precursors for LCs and inflammatory tumor necrosis factor- α (TNF- α) inducible nitric oxide synthase (iNOS)-producing ‘Tip’ DCs.

(MDPs),^{45,46} which, in turn, probably derive from common myeloid progenitors (CMPs). In two landmark papers,^{40,41} separate Ly6C^{hi} and Ly6C^{lo} monocytes were described that correspond to human CD14⁺ and CD16⁺CD14⁺ monocytes, respectively. In both mice and humans, monocytes have the capacity to differentiate into macrophages in response to M-CSF⁴⁷ and into CD11c⁺ MHCII⁺ DCs in the presence of GM-CSF, with or without interleukin (IL)-4.^{7,48–55} This DC transition can occur for both monocyte subsets *in vitro*, and under some conditions *in vivo*.^{40,41,56–58}

As such, monocytes, or their earlier myeloid precursors, were thought of as the definitive pre-DCs, and the DCs derived *in vitro* with GM-CSF as representative of those found normally in a mouse or patient. However, transfer of Ly6C^{hi} monocytes into steady-state or

irradiated mice were never able to give appreciable DC generation,^{40,41,46,59} except for two studies where low-level generation was observed.^{57,60} Transfer of the resident Ly6C^{lo} subtype could produce some DCs in spleen, lung and lamina propria.^{46,60,61} However, in some of these cases, inflammation or the experimental clearance of DCs was required for efficient engraftment. Whether these latter situations represent an ‘emergency’ process, or simply amplify the normal process, is not clear. The precise role of monocytes in DC generation *in vitro* and *in vivo* is discussed later.

Plasmacytoids

pDCs⁶² are another type of pre-DCs in the steady state and are found in many tissues of the mouse, including blood, thymus, BM, liver and

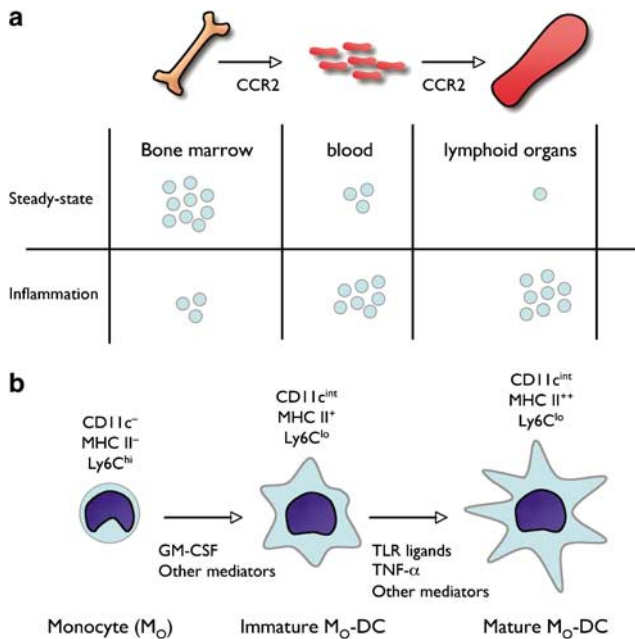


Figure 2 The *in vivo* pathways of Ly6C^{hi} monocyte-derived dendritic cell (DC) development. **(a)** In the steady state, bone marrow (BM) is a reserve for monocytes and low levels circulate in a surveillance role. Upon infection or inflammation, monocytes are released from the BM via a CCR2-dependent to enter the blood and lymphoid organs. **(b)** Upon such a stimulus *in vivo*, monocytes can convert to DCs as a result of granulocyte-macrophage colony-stimulating factor (GM-CSF) and/or other mediators. Further activation of DCs might require direct contact with pathogens or high levels of tumor necrosis factor- α (TNF- α) or other mediators in the microenvironment.

lymphoid organs^{37,63–69} (Figure 1). Their best-described role is in pathogen surveillance where, upon exposure to viruses,⁷⁰ bacteria^{65,71} and certain TLR agonists^{72–75} they produce a host of inflammatory chemokines⁷⁶ and cytokines, including a characteristic type I interferon (IFN) burst. IFN- α production by pDC has been ascribed as an essential ‘third signal’ for the generation of CD8 T-cell memory, in addition to the two signals proved by cDC peptide-MHC and co-stimulation.^{77,78} pDCs also have an overlapping role as APCs, and express low levels of MHC I, MHC II and co-stimulatory molecules in the steady state, all of which are upregulated upon activation.^{68,79} At such time, pDCs concomitantly acquire dendritic processes and are thus instilled, in some cases^{80,81} but not all, with the potential for T-cell activation. Collectively, these recent studies of pDCs enlist them as crucial mediators linking the innate and adaptive arms of the immune system.

As pDCs are found in most organs of the body, the location of pDC precursors has been difficult to pinpoint.⁸² They appear to develop in the BM and/or blood, then traffic to the lymphoid organs via the blood as developed pDCs, similar to B cells. Evidence for this comes from the observations that developed pDCs are found in significant numbers in the blood and BM,^{66,70,83} that pDC circulation is CD62L (L-selectin) dependent^{67,84,85} akin to T and B cells and that their immediate precursors (CD11c^{int}B220⁺Ly49Q⁺MHCII⁺) are not detected in the peripheral lymphoid organs, whereas developed pDCs (CD11c^{int}B220⁺Ly49Q⁺MHCII⁺) are.^{86–89} They can be derived from either *fms*-like tyrosine kinase 3 (Flt3)-expressing CMP or common lymphoid progenitors (CLP) and, independent of their derivation, can also have features that overlap with lymphocytes.^{83,90–93} More recent progress in pDC development is discussed later.

Langerhans cells

LCs are stellate DCs that make up 2–4% of total cellularity in the epidermis.⁹⁴ Despite the initial excitement of LCs having an immune activating role^{95–102} and the popular dogma that LCs are the T-cell priming DCs for cutaneous infection, *in vivo* evidence is still lacking. In fact, two studies demonstrate that they are not involved in T-cell priming for some viral infections.^{103,104} Another interesting study points to a role for LCs in limiting HIV transmission¹⁰⁵ by using Langerin and their unique tennis racket-shaped Birbeck granules.^{106,107} as a sort of ‘trap’. Some evidence even points to a role in T-cell tolerance to skin antigens or commensal bacteria.^{20,108–111} Despite these recent insights, the full panel of LC function is still lacking.

The origin of LCs is still mired in controversy. It is known that they are a long-lived population in the epidermis and constitutively migrate from the skin to the draining lymph nodes^{24,25,112–114} and that 25–60% turn over every 2 weeks.²¹ But ‘turning over’ does not necessarily mean derivation from earlier precursors.

Initial suggestions of a blood-derived precursor came from mice analysed at late time points after irradiation.^{115–117} Other studies instead suggested LCs either proliferated themselves or could develop from skin-resident precursors.^{118,119} Studies from Merad and colleagues have convincingly shown that both can occur, depending on the situation.^{120,121} Only upon exposure to UV light or skin damage are LCs depleted from skin and reconstituted from circulating precursors in a CCR2-dependent process¹²⁰ that recruits CCR2-expressing Ly6C^{hi} monocytes.⁵⁶ However, in the absence of inflammation, LCs homeostatically divide or derive from local precursors.

A monocyte origin of LCs in these circumstances fits with prior evidence, that LCs are members of the MPS^{122–124} that *op/op* mice (deficient for M-CSF and, thus, monocytes) have reduced LCs,¹²⁵ and that LCs can be derived from monocytes *in vitro* with GM-CSF $\pm$ TGF- β .^{26,126,127} More recent findings in LC biology, including evidence that backs the suggestion that LC homeostasis might be established early during ontogeny,⁷ can be found in the most recent LC meeting highlights.¹²⁸

Interstitial DCs

Interstitial DCs are found throughout the body and comprise those found in all peripheral tissues, excluding the LCs of the epidermis. They include the well-characterized dermal DCs of the skin¹²⁹ and DCs of the mucosae.^{130,131} These DCs share some characteristics with LCs but also have many differences.^{132–134} An example of their similarity is that interstitial DCs can also migrate, even in the steady-state and germ-free situations, to draining lymphoid organs where they appear as CD11c^{int} MHC II^{hi} DEC-205^{int} DCs^{21,24} (Figure 1). Although there are indications that interstitial DCs are an immune-regulating APC based on response to injected antigen or hypersensitivity responses,^{135,136} a role for interstitial DCs during an actual infection has only been demonstrated after infection with intravaginal herpes simplex virus-2 (HSV-2)¹⁰⁴ and *Leishmania*.^{137,138} Dermal DCs also appear to be derived locally from as yet unknown precursors.¹³⁹ In addition to dermal DCs, dermal macrophages and LCs of the epidermis, several groups have identified other langerin-expressing DCs,^{121,140,141} although a clear differential function for these is still being investigated.

Resident CD8⁺ DCs

CD8⁺ cDCs have been the focus of several exciting studies over the last decade. Unlike other resident cDCs, they are found characteristically in the T-cell areas of the spleen and lymph nodes in the steady state

(Figure 1),¹⁴² and probably correspond to the interdigitating cell that had been described previously.^{4,143,144}

One special role for CD8⁺ cDCs might be in the maintenance of tolerance to self. As not all autoreactive T cells are deleted in the thymus during negative selection,¹⁴⁵ there is need for a peripheral mechanism of self-tolerance.¹⁴⁶ Peripheral tolerance is mediated by a hematopoietic cell^{147–149} and can be DC-mediated.^{150,151} The proximity of CD8⁺ cDC to T cells in the steady state place them in prime position as the responsible APC.¹⁵² Further evidence came from tissue-specific expression of model antigens for which only CD8⁺ DC could present antigen (implying tolerance),¹⁵³ and in the diabetes-prone nonobese diabetic (NOD) mouse, where there was a positive correlation of CD8⁺ DC numbers and the reversal of diabetes.^{154,155}

As DCs were once thought only to be capable of presenting endogenous antigen via MHC I, and exogenous antigen via MHC II, the question arose of how exogenous self-antigen could access the MHC I pathway of DCs for the induction of tolerance or for immunity. One of the unique key features of CD8⁺ DCs is their ability to ‘cross-present’ antigen, which occurs when exogenous antigen is diverted to the MHC I pathway and thus able to signal the TCR of CD8⁺ T cells.^{156–158} This process is important for both peripheral T-cell tolerance and other scenarios,¹⁵⁹ although the precise molecular mechanisms are still ill-defined and controversial.^{160,161}

In apparent contradiction to their role in T-cell tolerance, CD8⁺ DCs also seem to be the main T cell-priming APC in several viral and bacterial infections^{103,156,157,162,163} and the targets of ‘licensing’, whereby CD4⁺ T cells activate DCs, which, in turn, cross-present antigen to induce memory CD8⁺ T-cell generation.¹⁶⁴ How can the one DC subtype perform the dual yet opposing roles of immunity and tolerance? A differential activation state and cytokine profile within the one CD8⁺ cDC population might be the factor. However, there is still the formal possibility that there are ‘immunogenic’ and ‘tolerogenic’ subtypes of CD8⁺ DCs, although no data to date support this view.

CD8⁺ cDCs are the only DCs to express TLR3 and fail to express TLR7.¹⁶⁵ In this way, CD8⁺ cDCs may be specialized for dsRNA but not ssRNA viruses.^{166,167} Activation of CD8⁺ cDCs induces upregulation of MHC II and co-stimulatory molecules, as well as large amounts of ‘bioactive’ IL-12p70.^{168,169} This cytokine is crucial for CD4⁺ Th1, cytotoxic CD8⁺ T-cell priming and memory generation.^{170,171} Therefore, TLR or other danger signals probably provide the switch for CD8⁺ DCs from a tolerogenic to an immunogenic state.

Resident CD8[−] cDC

CD8[−] cDCs are the most numerous of the resident cDCs in spleen, comprising roughly 80% of the total, but are a minor population in lymph nodes. In contrast to CD8⁺ cDCs in the T-cell areas, CD8[−] cDCs tend to be found in the marginal zones,^{142,143,172} which appears to be dependent on the CCR6/MIP-3 α axis.^{173,174} However, upon stimulation by lipopolysaccharide (LPS) or other TLR agonists, CD8[−] cDCs relocate to the T-cell areas of lymphoid organs.^{142,168} They also secrete large amounts of the inflammatory chemokine RANTES, MIP-1 $\alpha\beta$ and MIP-1 β .⁷⁶ Despite a recent report of their potential importance in yeast infection,¹⁷⁵ other infections where they play an important role is yet to be determined.

GM-CSF-DERIVED DCs AND INFLAMMATORY TIP DCs: ONE AND THE SAME?

There is now overwhelming evidence that monocytes are not the major precursors for steady-state DCs, with some exceptions (see

Table 2 *In vivo* situations where inflammatory Tip DCs emerge

	References
<i>Parasites</i>	178
<i>Leishmania major</i>	
<i>Bacteria</i>	
<i>Salmonella typhimurium</i>	179
<i>Streptococcus pneumoniae</i>	180
<i>Brucella menitensis</i>	181
<i>Listeria monocytogenes</i>	43
<i>Mycobacterium tuberculosis</i>	182,183
<i>Escherichia coli</i>	184
Commensal bacteria	185
<i>Viruses</i>	
LCMV	186
Influenza	187
<i>Autoimmune</i>	
Diabetes	188
Psoriasis	189

Abbreviation: LCMV, lymphocytic choriomeningitis virus.
With some adaptation from Wick.¹⁹⁰

above). Instead, they readily convert to DCs in inflammatory situations^{40,58–60,176,177} (Figure 1; Table 2). Related to this is that the DCs generated when monocytes or earlier progenitors are cultured with GM-CSF represent these inflammatory scenarios.¹⁹¹ Considering this culture method is the most widely used protocol for DC generation when assessing genetic, synthetic, biological or pathogenic factors in DC biology, the interpretation of numerous GM-CSF-derived DC studies that claim to represent the steady-state situation may need to be revisited.

GM-CSF-derived DCs: not necessarily representative of the steady state

As monocytes generate a relatively homogeneous population of CD11c⁺CD11b⁺CD8[−]MHC II⁺ ‘myeloid’ DCs in GM-CSF cultures, they have often been referred to as the equivalent of CD8[−] cDCs of the spleen and lymphoid organs, which share this pattern of marker expression. However, formal proof of such a correlation is lacking. Indeed, several lines of evidence suggest monocyte-derived DCs are not related to the development of steady-state DCs, apart from LCs and interstitial DCs in some situations.

First, it is sometimes underrecognized that GM-CSF levels are ‘below detection’ in steady-state mice and in healthy humans^{192–197} and so may not contribute to significant steady-state *in vivo* DC generation from monocytes. There are three possibilities that could count for this discrepancy between the presence of DCs *in vivo* and levels of GM-CSF; GM-CSF levels, although low, are still functional; there are factors *in vivo* that substitute for GM-CSF to generate DCs; steady-state DCs may actually be a distinct pathway from those that are monocyte derived.

Second, knocking out the genes of GM-CSF or its receptor in hematopoietic cells would never allow DC generation if this cytokine was obligatory. However, knockouts for either gene have normal ratios and number of the splenic resident DC subtypes, are present in their expected ratio and number, and with no difference in function.^{198,199} M-CSF null mice (*op/op*) and M-CSF receptor^{−/−} have severely

perturbed monocyte development in early life and an absence of some macrophage subsets.^{56,200–202} Although total DC numbers are down in these mice,²⁰³ it is in accordance with total spleen size and so probably does not reflect a specific defect in the DC compartment (Witmer-Pack *et al.*²⁰⁴; D Vremec, personal communication). Of the DCs that are present, the splenic subtypes are also in their expected ratio and density and exhibit normal functions (in the assays tested), suggesting M-CSF is not obligatory for steady-state DC precursors, nor are the precursors monocytes. However, M-CSF might play a role in the maintenance of the pool of DCs¹⁸ and in augmenting DC development.²⁰⁵ M-CSF has also recently been described as a novel DC poietin. In that study, the nonadherent cells ordinarily discarded in M-CSF cultures (which are employed to generate macrophages) contained DCs.²⁰⁶

Third, the major cytokine driving the generation of the steady-state cDC subtypes *in vivo* appears to be Flt3 ligand (FL) as evidenced by an absence of DCs in genetic knockouts of FL, or members of it signaling pathway, such as STAT3.^{207,208} By contrast, these mice have normal monocyte development²⁰⁹ and GM-CSF-derived DC development.²⁰⁷

Moreover, it is important to note that *in vivo* CD8[−] DCs and GM-CSF-derived DCs are fundamentally different in terms of their derivation from precursors,^{59,210} in their function,¹⁹¹ and in their genetic makeup.²¹¹ Thus, despite popular assumption, they may represent distinct DC types.

Inflammatory TNF- α iNOS-producing DC

When would GM-CSF become important for monocyte-derived DC generation *in vivo* if it is not present at significant levels in the steady state? Infections such as *Listeria*¹⁹⁵ and other situations induce higher serum levels of the cytokine.²¹² In fact, GM-CSF was discovered using lung preparations of mice treated with LPS—certainly not a steady-state scenario.²¹³ The first clear hint of the *in vivo* equivalent of GM-CSF-driven monocyte-derived DCs came when a novel DC subtype was identified in the spleen of mice 1–2 days after infection with *Listeria monocytogenes*.⁴³ This TNF- α (tumor necrosis factor- α) and iNOS (inducible nitric oxide synthase) producing DC (Tip DC) was unique in phenotype in that they were CD11c^{int}CD11b^{hi}MAC-3⁺ unlike splenic steady-state cDC, which are CD11c^{hi}CD11b^{low}MAC-3[−]. In mice deficient for the chemokine receptor CCR2, this DC subset failed to develop in spleen, but CD11c^{hi} cDC numbers were normal. Interestingly, the Ly6C^{hi} monocyte subset selectively expresses CCR2,⁴⁰ and so was presumed to be the contributing precursor to this kind of DC.⁴⁴ Direct *in vivo* evidence for such a process came in our model of inflammation where CD11c[−]MHCII[−]Ly6C^{hi} monocytes developed into CD11c^{int}MHCII⁺Mac3⁺ DCs.⁵⁹ Since the description of Tip DCs, and even before, other CD11c^{int}CD11b^{hi} DCs have been described that arise during many bacterial, viral and parasitic infections, and autoimmune sequelae (summarized in Table 2). Whether these are involved in immunity, infection control or regulation will be very important to dissect in the future.

GM-CSF-derived DCs are the *in vitro* equivalents of Tip DCs

In light of the above evidence, Xu and colleagues explored whether GM-CSF-derived DC cultures represent Tip DCs rather than those of the steady state.¹⁹¹ By all parameters tested, this appeared to be the case. Specifically, GM-CSF-derived DCs, but not FL-derived DCs (see below), were large and granular, had the phenotype CD11c⁺MHCII⁺Mac-3⁺, could produce TNF- α and iNOS in response to TLR agonists, and could develop from Ly6C^{hi} monocytes—all parameters which aligned them with Tip DCs, and not those of the steady state or from FL cultures. Interestingly only activated GM-CSF-

derived DCs and Tip DCs produced chemokine ligand 2 (CCL2), perhaps pointing to roles in further monocyte recruitment (CCL2 binds CCR2 on Ly6C^{hi} monocytes), and produced IL-10, perhaps indicating an immunosuppressive role. Related to this, Ly6C^{hi} monocytes have been implicated as the myeloid suppressor cells in a mouse model of multiple sclerosis.²¹⁴

FL-DERIVED DCs REPRESENT STEADY-STATE CD8[−], CD8⁺ AND pDCs

If GM-CSF cultures do not best represent the steady-state DC pathway, then can any other culture methods substitute? Immunex scientists provided the answer in a key publication,²¹⁵ with a more thorough examination of the question from our laboratory.¹⁷⁴ In view of the observations that FL treatment increased DC numbers,^{216,217} and that FL^{−/−} mice had abrogated DC numbers,²⁰⁸ the Immunex group devised a novel culture system to generate DCs.²¹⁵ BM cell cultures supplemented with FL for 9 days generated DCs to roughly the same number as total BM cells initially seeded. Interestingly, and in an often-overlooked paper, these DCs were morphologically and phenotypically distinct from GM-CSF-derived DCs,²¹⁸ with further characterization of these differences by Xu and colleagues¹⁹¹ (see above).

Despite the lack of CD8 or CD4 expression, the investigators^{215,219,220} classified three DC subtypes in these cultures; CD11c⁺B220⁺ pDCs, CD11c^{hi}CD11b^{lo} DCs and CD11c^{lo}CD11b^{hi} DCs. Considering CD11c^{hi}CD11b^{lo} DCs (which would have also contained pDCs) upregulated surface CD8 after LPS activation and secreted IL-12 p70 upon activation, they were proposed to be similar to CD8⁺ cDC of the spleen.

Using new markers, we better separated two cDC subtypes into CD24^{hi}Sirp- α [−] and CD24^{lo}Sirp- α ⁺ DCs, and demonstrated they were indeed *bona fide* equivalents of CD8⁺ and CD8[−] cDCs—not only phenotypically, but also functionally (Figure 1). For example, only the CD8⁺ FL-DC equivalents expressed cystatin C and TLR3, produced IL-12 p70 in response to TLR agonists, could cross-present cell-associated antigen to CD8⁺ T cells, and were dependent on the interferon regulatory factor (IRF-8) transcription factor for its development, among other unique features.¹⁷⁴ Worth noting here for those wishing to use this system is that CD8⁺ FL-DC equivalents could not cross-present soluble antigen, unlike their *in vivo* counterparts (SH Naik and NS Wilson, 2004, unpublished observation).

If the research purpose is to investigate the development or function of the steady-state pDC, CD8⁺ or CD8[−] cDCs, then FL cultures are currently the simplest and most representative *in vitro* protocol. Most importantly, one can get up to 20 times more total DCs from BM FL cultures per mouse compared to spleen, and up to 80 times more CD8⁺ DC equivalents. Access to a mouse FL-expressing Chinese Hamster Ovary cell line is available upon request.

THE STEPS IN DC DEVELOPMENT

Determined so far...

So what are the steps in DC development? Our first line of experiments to recapitulate the findings of others^{57,221} that Ly6C^{hi} monocytes and CD8[−] cDC could be precursors of CD8⁺ cDC did not yield the same results.^{59,222} This fitted with previous and subsequent data^{40,41,46,56} which destroyed the age-old assumption that monocytes were precursors of all DCs. Instead, we and another group identified spleen-resident precursors that were not monocytes but that could generate DCs upon transfer with relatively high efficiency into non-irradiated recipients.^{39,59} These we termed pre-cDC. Although representing a minor population within spleen (0.05%), they accounted for

most of the spleen's DC precursor activity in a 5-day *in vivo* transfer assay, could generate both DC subtypes without much further division and could even be further separated using CD24 into CD8⁺ cDC-only precursors or CD8⁻ cDC-only precursors (Figure 1). The equivalents of CD11c⁺MHCII⁻ pre-DCs were also found in BM and other lymphoid organs,^{39,89} and at earlier time points in FL cultures.²¹⁰ Thus, DCs can arise from a precursor that is resident within the lymphoid organ itself, and this cell is not a monocyte.

What about earlier precursors to the pre-DCs? As expected, DCs can be derived from hematopoietic stem cells (HSCs), but they are unique in their derivation from both CMPs and CLP progenitors, as long as they express surface flt3.^{223–227} Further downstream, Fogg *et al.*⁴⁵ have identified a precursor within the CMP fraction that could produce cDCs (both CD8⁺ and CD8⁻) and macrophages, but not granulocytes, pDCs or any other lineage (Figure 1). In contrast, our group²¹⁰ and that of Manz²⁰⁵ have identified a precursor in BM as well as in FL cultures, termed pro-DCs, which could generate both CD8⁺ and CD8⁻ cDC subtypes as well as pDCs, but not macrophages or any other lineage (Figures 1 and 3). Pro-DCs generate pre-DCs *en route* to the generation of the DC subtypes.

Whether MDP and pro-DCs represent distinct or overlapping populations is not clear (Figure 3). What might account for the differences? Phenotypically, MDPs and pro-DCs are very similar in phenotype (both are Lin⁻Sca-1⁺IL7R⁻CD117^{int}). However, at least in our hands, most but not all pro-DCs are CD34⁻CD16/32⁻, whereas MDPs are largely CD34⁺CD16/32⁺. We also find that not all *in vivo* pro-DCs are M-CSFR⁺.¹⁹⁷ A second important difference is that the majority of pro-DCs do not generate colonies in response to GM-CSF, whereas MDP do. Considering GM-CSF-derived DCs better represent inflammatory DCs (above), and clonal studies of MDP used this cytokine, it is possible that MDPs are precursors of monocyte-derived DCs, but not monocyte-independent steady-state DCs. Arguing against this was that MDPs could generate large numbers of CD8⁺ and CD8⁻ cDCs, but no pDCs, on a per-cell basis *in vivo*.⁴⁵ However, the assays in that part of the study were not clonal, so there is a possibility that macrophage and DC precursors were distinct populations within cells classed as MDPs. Another possibility is that there are

multiple routes to DC production and both pro-DCs and MDPs make a contribution. There is precedence for this considering both CLPs and CMPs can give rise to all splenic DC subtypes. Certainly further comparison between the two progenitors will be required.

Blood-derived or organ-resident precursors

Since DCs are found in the blood of humans, it had been generally assumed that they are derived in the BM and circulate as DCs to the organs, and there is evidence to support this.^{35,228,229} However, as several groups including ours have identified or implied organ-resident DC precursors,^{39,56,59,89,139,210} it suggests that the final step of differentiation may take place within the organ itself for some DC subtypes. We speculated in one study that the existence of pre-cDCs could indicate derivation from a long-term organ-resident reservoir of DC precursors, and/or that they represented 'in-transit' cells from the BM *en route* to DC development.⁵⁹ Liu *et al.*²³⁰ tackled this question elegantly by utilizing mouse pairs that differ in CD45 allotype and are physically joined, like Siamese twins, so their bloodstream and therefore leukocyte migration is shared (parabiotic mice). As lymphocytes are circulating and long lived, they reached a 50:50 host:partner chimerism 7 days after anastomoses, as would be expected. However, only 17–35% of host DCs were partner derived. The authors concluded that this incomplete DC chimerism reflected the short half-life of DC precursors in blood—that is, most host-derived blood-borne DC precursors have already died or extravasated host tissue before they can reach the partner. Formally, the parabiosis experiments cannot fully exclude longer-term intrasplenic DC precursors for the remaining 15–33% of host-derived DCs. However, their most convincing data that a major proportion of splenic DCs derive from circulating DC precursors was from the tracking of DC origin after parabiosis separation, where most DCs were again of host origin within 10–14 days. Caution should be exercised when interpreting this as a longer half-life of DCs, rather than the 3–5 days originally described,^{231,232} if there were remnant donor DC precursors also present in partner spleen after separation. Nevertheless, unless the process of parabiosis separation somehow alters DC homeostasis, it appears that many, if not all, DCs are derived from circulating precursors.

What then are such circulating DC precursors? One or more of many stages of the DC development process could account for replenishment. DCs themselves may develop in BM and then be released into the bloodstream to replenish the splenic DC pool, even though such cells are barely detectable in the bloodstream.³⁷ Alternatively, one or more of the pre-DC,^{39,59,89,210} pro-DC,^{205,210} MDP⁴⁵ or an as yet unidentified DC precursor may do the job. Pre-DCs are clearly a good candidate considering they are found in many organs, including BM, blood and spleen, and can give rise to DCs.^{39,59,89,210} Cells with the phenotype of pro-DCs are also present in spleen (SH Naik, unpublished observations), so may also play a role.

None of the aforementioned studies need be contradictory—just their interpretation. Thus DC homeostasis may involve a coordinated regulation of limited to major self-renewal^{156,139,233,234} and derivation from blood-derived precursors²³⁰ that include one more of the pre-DC, pro-DC or MDP populations.

THE 'GRADED' COMMITMENT MODEL

The large expansion of pro-DCs and the use of the FL culture system afforded us the unique opportunity to perform single cell, or clonal, assays for the first time. We asked the simple question of whether a single precursor can produce all three of the pDC, CD8⁺ and CD8⁻ cDC subtypes, or whether there are individual subtype-committed

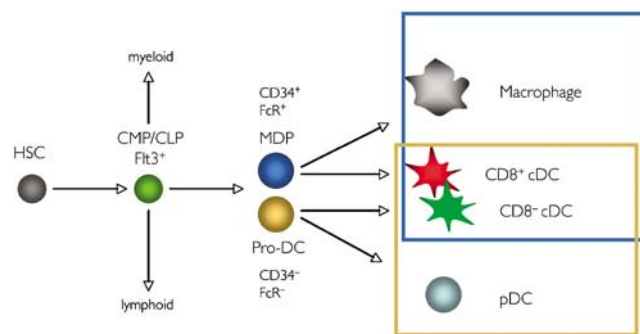


Figure 3 The confusion regarding dendritic cell (DC) development from 'intermediate' stage precursors. All DCs appear to derive from *fms*-like tyrosine kinase 3 (flt3)-expressing progenitors, whether they are common myeloid progenitor (CMP) or common lymphoid progenitor (CLP) derived. One group has identified a downstream macrophage-DC precursor (MDP) of conventional CD8⁺ & CD8⁻ conventional DCs (cDCs) and macrophages, but not plasmacytoid cells (pDCs) or other lineages. In contrast, two other groups have identified pro-DCs (also known as common DC precursors (CDPs)) that generate conventional CD8⁺ and CD8⁻ cDCs and pDCs but not macrophages or other lineages. Whether these are distinct, overlapping or incorrectly described pathways remains to be determined.

precursors. Surprisingly, the answer was a mix of the two options. In fact, we found that single cells within what we classed as pro-DCs could sometimes generate one, sometimes two, and sometimes all three DC subtypes, with a predictable pattern between experiments. This could have several explanations. The first is that pro-DCs were not a 'pure' population and that there were still distinct populations within what we classed as pro-DCs, which our panel of tested markers could not discriminate. Alternatively, our pro-DC population was, in fact, 'homogeneous' and single cells can give rise to different outcomes on a cell-to-cell basis. For this latter option, stochastic cell-intrinsic cues, or signals from neighboring cells in culture, or both, could determine the cell's fate. Which of the above scenarios accounts for the phenomenon is not clear, but is currently under investigation. Moreover, whether such a process applies *in vivo* will also require further investigation.

In an attempt to find an upstream precursor stage to pro-DCs, where all single cells putatively generated all DC subtypes, we cultured single cells of still earlier multipotent progenitors (MPPs) that are capable of generating all hematopoietic lineages (MPPs are the major cell type within lineage⁻ckit⁺Sca-1⁺ BM cells). Although there was a greater proportion of clones that generated all three DC subtypes, to our surprise there were still single cells that could produce only one or only two subtypes. In particular, the percentage of pDC-only clones produced by MPPs was larger, as was their clone size, compared to pro-DCs. It is actually for this reason that we prefer the term 'pro-DCs' to 'CDPs' (common DC precursor), as the term 'common' implies that all cells can equally produce all DC subtypes, whereas pro-DC implies a cell 'in favor of' DC generation.

If these observations hold true *in vivo* of a 'propensity' of single cells to generate different DC subtype combinations at different stages of hematopoiesis, rather than an all-or-nothing event at a fixed stage, the classic branching models of hematopoiesis are not informative enough. For example, our data suggest that DC development in general can branch at many steps along the hematopoietic pathway with a more defined branch at the pro-DC stage. However, if one looks at pDC-only precursors, for example, these can branch off with a greater probability earlier in hematopoiesis, such as at the MPP stage, with some potential still remaining in downstream steps, such as the pro-DC step. For want of a better term, I will call this graded commitment—that is, each division down certain pathways has a probability of a committed precursor for a cell type/s 'branching off' (Figure 4), and this may increase (or decrease) on a population level depending on the lineage. Thus, perhaps part way down the hematopoietic pathway a single cell initiates the 'all DC program', which can then potentially switch to either the 'pDC program', 'CD8⁺ and CD8⁻cDC program', 'CD8⁺ DC program' and so on.

The molecular mechanisms that would guide such fates may include the numerous transcription factors that have been described that differentially guide DC subtype development (reviewed in Shortman and Naik⁷). Their regulation may be stochastic within the cell, or depend on external cytokine or cell-associated signals. The Flt3/FL axis may be an 'instructive' mediator in the 'all DC' program with evidence that enforced expression of committed Flt3⁻ megakaryocyte/erythroid progenitors could divert DC development.^{235,236} Another key paper that could support this model was published recently using human DCs that develop in NOD-severe combined immunodeficiency mice, which suggested the pDC and cDC development program is independent of the conventional myeloid and lymphoid pathways. The authors found that, no matter whether the DCs were CMP- or CLP derived, they had near identical gene expression profiles.²³⁷ This evidence can fit the graded commitment model such

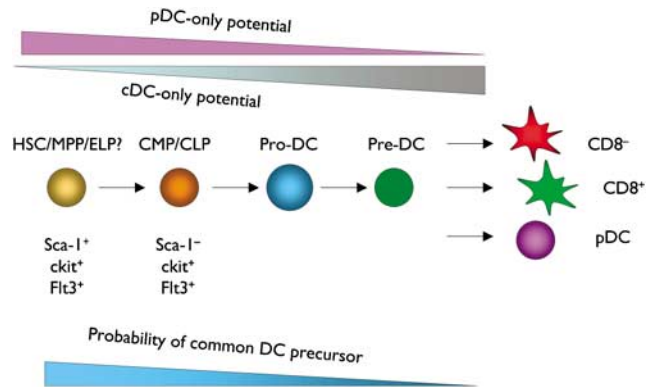


Figure 4 The 'graded' commitment model of dendritic cell (DC) development. There does not appear to be defined steps where every cell within a population can, or cannot make a particular DC subtype. Rather, there is a gradation of probability of getting an 'all DC' clone versus a plasmacytoid cell (pDC)-only or conventional DCs (cDC)-only clone from a single cell. The earlier in the pathway, the more likely there are single cells that can make not only 'all DCs' but also 'pDC-only'. Further downstream, these events are less likely, but the chance of cDC-committed clone increases. This model aims to explain how DC subtypes can derive from different progenitors in the pathway, with different patterns of clonal development on the single cell level. The molecular mechanisms that dictate this may involve the interplay of Flt3 ligand (FL) signaling, various transcription factors and the niche.

Table 3 Comparison of the mouse and human DC subtypes

	Mouse	Human	References
Plasmacytoid	pDC	pDC	239,240
Inflammatory Mo	Ly6C ^{hi} Gr1 ⁺	CD14 ⁺	40
Noninflammatory Mo	Ly6C ^{lo}	CD16 ⁺ CD14 ^{lo}	40
Langerhans cells	CD1a ⁺ LC	Langerin ⁺	241
T-cell area resident DC	CD8 ⁺ cDC	BDCA3 ⁺ DC (?)	211,242
Marginal zone resident DC	CD8 ⁻ CD4 ⁺	BDCA1 ⁺ CD4 ⁺ (?)	5,211,243

Abbreviations: Mo, monocytes; pre-DC, precursors of DC.

that a fixed 'DC program' can be initiated in single cells, even as far down 'opposite ends' of hematopoiesis as the lymphoid and myeloid pathways. Kincade and colleagues also provide important arguments against the classical binary models of hematopoiesis,²³⁸ which apply here. Although not discussed further, existing data for many cell types can fit such a phenomenon. An assertion of the relevance of this model *in vivo* would ultimately require good cell-tracing technologies.

ARE BDCA3⁺ HUMAN DCS EQUIVALENT TO MOUSE CD8⁺ DCS?

Correlating the human and mouse DC subtypes has been challenging. This has been owed largely to poor accessibility to human spleen and lymphoid tissue, the absence of key murine DC subtype markers on human DCs (and *vice versa*), and poor reagents that dissect the human DC subtypes. Some DC and monocyte subtype equivalents have already been identified, whereas others are not so clear (Table 3). Those of us whom work on the murine DC subtypes are quizzed incessantly about what the human equivalents of mouse CD8⁺ cDCs are, especially considering the vast array of tolerogenic and immuno-

genic functions attributed to this subtype in rodents.⁸ Alas, due to a chasm between murine and human DC research that is only now being bridged, there has never been a good answer. Here, an inadequately brief summary of human DC subtype findings and recent developments is presented.

As mentioned earlier, CD8⁺ DCs are barely detectable in the blood of a steady-state mouse and are instead found in the T-cell areas of lymphoid organs. As researchers of human DCs have generally focused on blood DCs or blood precursor derived (with GM-CSF) DCs, it is perhaps no surprise that they have not yet been 'discovered' in our species. Moreover, as described earlier, GM-CSF-derived DCs are not the best model for dissection of the CD8⁺ DC equivalent.

Some early potential clues for a human CD8⁺ DC equivalent can be found when probing the description of (1) DC heterogeneity in human spleen,²⁴³ (2) an interdigitating DC (not an LC) in the T-cell areas of tonsillar tissue that was CD11c⁺ MHCII⁺ DRC-1[−] (R4/23[−]) and CD1a[−] (Leu-6[−])²⁴⁴ and (3) an IL-12 p70-producing CD11b[−]CD45RO[−] thymic DC subtype.²⁴⁵ Hart and colleagues also thoroughly investigated human DCs and identified several subtypes,²⁴⁶ including a CD13^{hi}CD45RO^{hi} population that shares some characteristics with mouse CD8⁺ DCs²⁴⁷ and BDCA3⁺ DCs (see below).

More recently, the new BDCA (blood DC antigen) markers (1–4) from Miltenyi²⁴⁸ have facilitated a clearer distinction of human DCs, and identified a rare BDCA3-expressing cell in blood that was not a pDC, LC or 'myeloid DC'^{18,246,248}—at that stage, no further speculation as to its function was explored. Two more recent publications, which may be recounted as the first guided correlations, provide some furtive but potentially profound clues that human BDCA3⁺ DCs are the equivalents of mouse CD8⁺ DCs.

In the first study, scientists at Amgen further characterized the BDCA3⁺ DC subtype that expanded in patients injected with FL, a treatment known to expand DCs in humans and especially CD8⁺ DCs in mice.^{216,249} Using phage display and the 'Immunex spirit' (acknowledgements in Galibert *et al.*²⁴²), they identified a reactive protein that recognized the antigen nectin-like protein 2 (or Nect2) not only on BDCA3⁺ human DCs, but more importantly also on CD8⁺ mouse spleen DCs. Like murine CD8⁺ DCs,²⁵⁰ they were also CD11c⁺MHCII⁺CD2[−]CD13⁺CD16/32^{−/lo}CD33⁺CD162^{high}CD11b[−]HLA-DO⁺ and in the T-cell areas of spleen.

In another landmark study, Robbins *et al.*²¹¹ went further and compared gene expression profiles within and between the mouse and human DC subtypes using microarrays, and found some striking patterns and close matches. Although not unexpected, they nicely demonstrated correlations between mouse versus human pDCs, and mouse versus human GM-CSF-derived DCs.²¹¹ They also demonstrated a hierarchical relationship within the species' DC subtypes; for example, murine CD8⁺ and CD8[−] DC subtypes were most related, followed by their relationship to pDC. The relationship of these DC subtypes, however, was relatively distant from the interrelated myeloid lineages including monocytes, monocyte-derived DCs (with GM-CSF), macrophages and granulocytes. This fits nicely with our findings of late-branching between the cDC subtypes, their proximity with the pDC pathway, but their distinction from the myeloid pathway (see above and references^{7,9,59,174,210}). Another exciting observation was that BDCA3⁺ human DCs and CD8⁺ mouse DCs showed a 'partial functional equivalence'. Many genes that were common had not yet been ascribed a function. The exceptions were TLR3, also identified as unique within BDCA3⁺ DC and CD8⁺ cDC-specific in prior studies,^{32,251} and Nect2, as described above. Undoubtedly, any future study that potentially correlates the unique functional attributes of

CD8⁺ DCs in the mouse to the BDCA3⁺ DCs in human will be a seminal publication.

WHERE TO NOW?

Like always, there are many unanswered questions, and novel technologies may help to address some of these. Currently, lineage determination *in vivo* involves the transfer of a cell population. Although single cell *in vivo* assays are possible but difficult, they, or a cell-tracing technology to mimic this, are ultimately required to create meaningful maps of developmental pathways. Also of importance will be the ability to selectively deplete individual subtypes *in vivo*. Whether through a diphtheria toxin-based system,²⁵² or other methods, such tools could dissect the importance of the subtypes in various immune reactions, including the generation of Th1, Th2, Th17 and T_{reg} generation, in the clearance of various pathogens and cancer, in their contribution to autoimmune pathology and in other physiological homeostases. In terms of understanding DC development from precursors, the pathways still under debate will, of course, need to be clarified. However, we must eventually move away from protocols that involve extracting, sorting and injecting precursors intravenously, and explore other models where the real contributions of progenitors in an unperturbed *in vivo* situation are at work. Although the picture emerging for the precise pathways in steady-state hematopoiesis are becoming clearer, the changes in emergency hematopoiesis and DC development that occur upon infection/inflammation is still in its infancy^{253–258} but will be important to understand. The molecular mechanisms and *in vivo* niches that dictate lineage fates in these differing scenarios will also need to be unraveled. Finally, by closing the chasm between mouse and human DC research for example, through the use of humanized mice, discovery of novel markers conserved interspecies, or the like, our accrued understanding in the field of immunology hopefully will also translate to novel therapeutic options, which are already well under way.

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